Sky-lightmaps not properly initialized

MC-170010 Lighting

Status	Resolved / Fixed
Affected	1.15.1, 1.15.2 Pre-release 2, 1.15.2, 20w06a, 20w20b, 20w21a (+3 more)
Fixed in	23w16a
Reported	2020-01-17

STEPS TO REPRODUCE

1. Launch Minecraft and open the attached world.
2. Confirm that the target subchunk has no lightmap and no lightmap above it, while each of its four horizontal neighboring subchunks has a lightmap.
3. Schedule a large number of lightmap creations and light updates so the lighting engine is delayed.
4. Before the lighting engine processes the queued updates, place a stone platform in the target subchunk.
5. Allow the lighting engine to process the queued lightmap creations and updates.
6. Observe that the subchunk containing the stone platform remains insufficiently skylit: light contributions from the sides are missing, while light is present only from blocks directly below the platform.

OBSERVED BEHAVIOUR

When a skylight map is created for a topmost chunk section, it is initialized with a skylight value of 0 and then scheduled for relighting. If another skylight map is created above it before the lighting engine processes the pending initialization, the original initialization is canceled. The lower map therefore remains at 0 instead of receiving the expected skylight value of 15. Relighting also propagates skylight from above only, so when the neighboring sections already have lightmaps, no skylight is pulled in from the sides. This leaves areas beneath the affected section darker than they should be, with only blocks directly below recently updated blocks receiving light because block updates check their six neighboring positions. The same initialization logic can also produce stale lighting after a lightmap is removed. When the new topmost map is initialized, an empty section may be set to skylight level 15 and that value may be propagated without marking affected sections dirty, leaving the lighting data correct while the corresponding chunk meshes still render the old, darker light level until they are otherwise rerendered.

EXPECTED BEHAVIOUR

Topmost skylight maps in already lit chunks should be initialized to a skylight value of 15, with skylight correctly propagated from all relevant neighboring directions. Creating or removing lightmaps should not leave sections dark, alter lighting without updating their rendering, or require later block updates to correct the result. Lighting should remain consistent and all affected sections should display the correct brightness immediately.

ENVIRONMENT

Minecraft Java Edition Versions: 1.15.1, 1.15.2 Pre-release 2, 1.15.2, 20w06a, 20w20b, 20w21a, 1.16.2, 21w08b, 1.17.1 (includes snapshots)

ORIGINAL DESCRIPTION

This is a direct continuation of

MC-148580

Resolved

. I post it as a new report since the old one already contains several similar bugs and I don't want it to digress too much, given that it's already closed for quite a while now.

The issue remains the same, namely that the topmost lightmap is not correctly initialized to a light value of 15, but rather to a value of 0.

net.minecraft.world.chunk.light.SkyLightStorage.java

protected ChunkNibbleArray createLightArray(long pos) {

ChunkNibbleArray chunkNibbleArray = (ChunkNibbleArray)this.lightArraysToAdd.get(pos);

```
if (chunkNibbleArray != null) {
return chunkNibbleArray;
} else {
long l = ChunkSectionPos.offset(pos, Direction.UP);
int i = ((SkyLightStorage.Data)this.lightArrays).topArraySectionY.get(ChunkSectionPos.withZeroZ(pos));
if (i != ((SkyLightStorage.Data)this.lightArrays).defaultTopArraySectionY && ChunkSectionPos.getY(l) < i) {
ChunkNibbleArray chunkNibbleArray2;
while((chunkNibbleArray2 = this.getLightArray(l, true)) == null) {
l = ChunkSectionPos.offset(l, Direction.UP);
}
return new ChunkNibbleArray((new ColumnChunkNibbleArray(chunkNibbleArray2, 0)).asByteArray());
} else {
return new ChunkNibbleArray();
}
}
}
```

The lightmap will then be registered for relighting inside onLightArrayCreated(...) and finally be relighted in updateLightArrays(...). There are however some problems with this initialization code:

- If the subchunk is non-empty, light is only propagated from above, but not from the sides. There are cases where this is not sufficient. However, this issue is completely shadowed by the next one.

- The initialization is completely skipped if another lightmap is added above it.

I attached a world producing wrong lighting using this second problem. Since I lag the lighting engine to spawn mutible blocks at once before the lighting engine can react, the provided world might not reliably work on some systems, but at least it does reliably work on mine. Would be nice if someone can confirm this on their system.

The setup is as follows:

- We have some subchunk that does not have a lightmap and no lightmap above, but all its 4 neighbors do have one.

- Schedule a

MANUAL RESULT

Version used	
Reproduced?	YES / NO / PARTIAL
Crash file	
Notes	
Tested by / date	

Live issue: https://bugs.mojang.com/browse/MC/issues/MC-170010